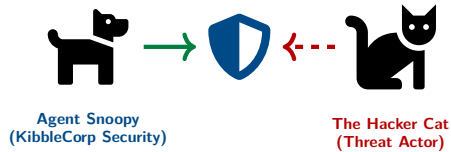


Module 9.0: Security Concepts

Intro to Networks

Brendan Shea, PhD

Rochester Community and Technical College



Outline I

Review: Operations and Monitoring Foundations

Key Module 8 Ideas

- Documentation and change control reduce operational risk.
- Monitoring data (SNMP, syslog, flow, packet capture) improves visibility.
- Baselines and alerts help teams detect and respond to anomalies quickly.

Why This Matters in Module 9

Security operations depend on reliable telemetry and disciplined operational processes.

Learning Outcomes I

After completing this module, you will be able to:

- Explain the **CIA Triad**: **Confidentiality** (data secrecy), **Integrity** (data accuracy), and **Availability** (reliable access).
- Distinguish between **authentication** (proving identity), **authorization** (granting permissions), and **accounting** (logging actions) in the AAA framework.
- Describe common threats: **DoS/DDoS** (denial of service), **malware** (viruses, worms, ransomware), and **insider threats** (malicious or negligent employees).
- Explain **social engineering** attacks including **phishing** (fake emails), **vishing** (voice phishing), **smishing** (SMS phishing), and **tailgating** (physical access).
- Describe **spoofing** attacks: **MAC spoofing** (impersonating devices), **IP spoofing** (forging source addresses), **ARP poisoning** (cache manipulation), and **DNS poisoning** (name resolution hijacking).

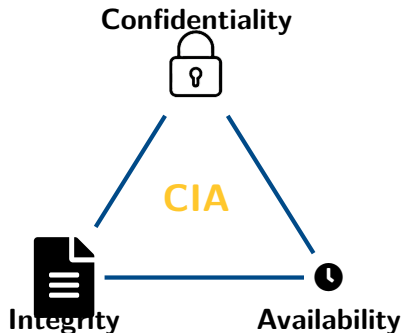
Learning Outcomes II

- Identify rogue devices: **rogue DHCP servers** (unauthorized IP assignment), **rogue access points** (evil twins), and explain mitigation using DHCP snooping and 802.1X authentication.
- Explain password attacks: **brute force** (trying all combinations), **dictionary attacks** (common words), **rainbow tables** (precomputed hashes), and defenses (complexity, length, salting, MFA).
- Describe **Defense in Depth** using layered controls: physical security, firewalls, IDS/IPS, access control lists (ACLs), encryption, and security awareness training.
- Use **Wireshark** to analyze packet captures and identify security indicators: DoS patterns, ARP poisoning, plaintext credentials, and suspicious traffic.
- Explain the role of **penetration testing**, **vulnerability scanning**, **honeypots** (decoy systems), and **SIEM** tools in proactive security operations.

9.1 Security Terminology: The CIA Triad

The Holy Trinity of Security

- **Confidentiality** (Secrecy)
 - Only authorized people can see data.
 - *Analogy*: A diary with a lock.
- **Integrity** (Accuracy)
 - Data hasn't been changed or tampered with.
 - *Analogy*: A sealed food jar (safety seal).
- **Availability** (Uptime)
 - Data is accessible when needed.
 - *Analogy*: A 24/7 convenience store.



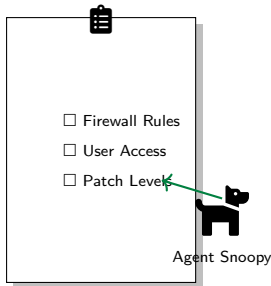
Security Audits

Why audit? To find broken locks before thieves do.

- **Internal:** Snoopy checking his own doors.
- **External:** Hiring a pro to break in (Pen Test).

Regulatory Compliance (The Law)

- **HIPAA** (Health): Protects *medical records* (e.g., your doctor can't tweet your X-ray).
- **GDPR** (Privacy): EU rules giving users control over data ("Right to be Forgotten").
- **PCI-DSS** (Finance): Rules for handling *Credit Cards*.



Encryption: Locking the Data

Symmetric (Private Key)

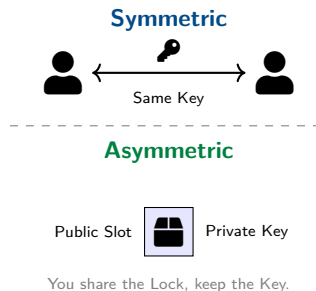
"The House Key"

- Uses **one shared key** for everything.
- *Pro*: Very fast (good for streaming video).
- *Con*: If you lose the key, you lose the house.

Asymmetric (Public Key)

"The Mailbox"

- **Public Key**: Anyone can put mail IN (Encrypt).
- **Private Key**: Only YOU can take mail OUT (Decrypt).
- Solves the key sharing problem!



Vulnerability and Exploit Types

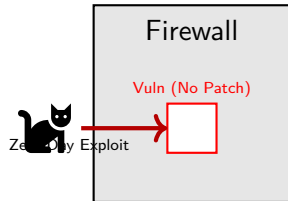
Definitions

- **Vulnerability:** A flaw (e.g., leaving a window open).
- **Exploit:** The tool to attack the flaw (e.g., a ladder).
- **Patch:** The board you nail over the window to fix it.

Zero-Day Attack

A vulnerability the vendor **doesn't know about yet**.

- **Days since fix = 0.**
- Extremely dangerous because there is no patch!



Defense Tool: Honeypots and Deception

What is a Honeypot?

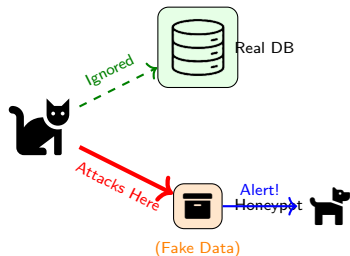
A "Trap Server" designed to look juicy to hackers.

- **Purpose:** To waste the attacker's time and study their methods.
- If anyone touches the Honeypot, **it's an attack** (because no real users should be there).

Common Tools

Canary: A device that chirps (alerts) when accessed.

Cowrie: A fake SSH terminal that logs the hacker's commands.



Example: Threat Modeling a Small Network

Threat modeling is just structured guessing: what do we care about, who might attack it, and how could they do it?

Asset	Likely Threat	Starter Control
Student accounts	Password reuse / phishing	MFA and password policy
Switch ports	Rogue device or evil twin	Port security and 802.1X
File server data	Ransomware or accidental deletion	Backups and least privilege
Internet access	DNS or DHCP spoofing	DHCP snooping and DNS hygiene

Beginner takeaway

Start with your most valuable assets, then ask what could break confidentiality, integrity, or availability.

9.2 Threat Types and Assessment

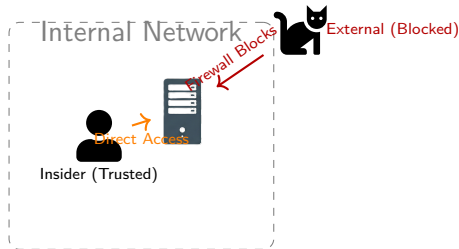
The Attacker Hierarchy

- **Script Kiddies:** Unskilled amateurs using tools they downloaded (Low threat).
- **APTs (Nation-States):** Highly skilled, government-funded spies (High threat).

The Insider Threat

Current or former employees.

- **Malicious:** Angry ex-admin.
- **Accidental:** Employee clicking a phishing link. (Most common!)



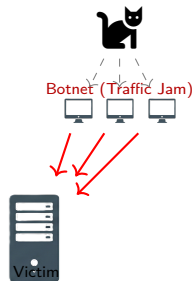
Denial of Service (DoS) vs. Distributed DoS (DDoS)

The "Traffic Jam" Analogy

- **DoS**: One car parks in front of the store entrance. (Annoying, but easy to tow/block).
- **DDoS**: Every car in the city drives to the store at once. (Catastrophic, impossible to clear quickly).

Goal

To attack **Availability**. The server isn't hacked, it's just overwhelmed.



Example: Security Controls Map

Beginners learn faster when a threat is paired with a control that actually reduces it.

Threat	Defensive Control	Why it helps
Spoofed packets	ACLs and filtering	Limits what forged traffic can reach
Password attacks	MFA and salting	Makes stolen or guessed passwords less useful
Rogue DHCP server	DHCP snooping	Blocks unauthorized address assignment
Ransomware	Backups and patching	Restores data and reduces infection paths
Social engineering	Training and verification	Slows down human manipulation

Botnets: The Zombie Army

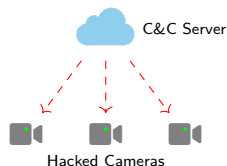
Who are the Zombies?

Often insecure **IoT (Internet of Things)** devices:

- Smart Cameras, Smart Bulbs, Connected Fridges.
- Weak passwords make them easy targets!

Structure

- 1 **Herder:** The criminal.
- 2 **C&C:** Server giving orders.
- 3 **Zombies:** The infected devices.



Malware: Virus vs. Worm

Virus (Needs a Driver)

Like a car that needs a driver, a Virus needs ****YOU**** to start it.

- It hides inside a legitimate file (game, document, script).
- It only infects when you **open/run** that file.



Worm (Self-Driving)

A Worm is a ****self-driving car****.

- It doesn't need a host file or a user.
- It scans the network, finds a hole in a neighbor's computer, and jumps over automatically.
- **Much faster** spreading than viruses.

Malware: Trojans & Ransomware

Trojan Horse (The Trick)

Named after the Greek myth. It looks like a gift, but contains soldiers (malware).

- *"Download this Screen Saver!"*
- Screen saver runs, but it opens a **Backdoor** for the hacker.

FreeGame.exe



Hidden Inside



File Encrypted!
Pay 5 BTC

Ransomware (The Kidnapper)

It doesn't steal data; it **locks** it.

- Encrypts your homework/photos.
- Demands Crypto (Bitcoin) to buy the key.
- *Advice:* Never pay! (Restore from Backup).

Case Study: The Zombie Accountants

Scenario: 2:00 AM Alert

Agent Snoopy's pager buzzes. The KibbleCorp webstore has crashed.
He checks the network dashboard:

- The computers in the Accounting Department are all turned ON (they should be off).
- They are sending thousands of requests per second to the webstore.
- The accountants are asleep at home.

Detective Questions:

- 1 Why did the malware likely spread as a **Worm** instead of a Virus?
- 2 The computers are acting as a group. What is this "Army" called?
- 3 Is the attack trying to steal data or stop the business?

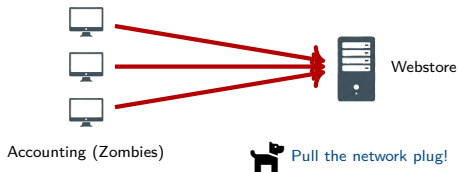


It's an inside job!

Case Study Solution: The Zombie Accountants

Snoopy's Analysis

- ➊ **Propagation:** It must be a **Worm** because it spread automatically while no one was using the computers.
- ➋ **The Army:** They are a **Botnet** (Robot Network) of Zombies.
- ➌ **Goal:** This is a **DDoS (Distributed Denial of Service)**. The goal is to overwhelm the webstore (Availability), not steal data (Confidentiality).



9.3 Spoofing: The Fake Return Address

What is Spoofing?

Falsifying data to gain an illegitimate advantage.

- **IP Spoofing:** Writing a fake "Return Address" on a packet.
- **Caller ID Spoofing:** Scammers calling from "Your Bank" (when they are really in a basement).



Why does this work?

The Internet (IP) is like the Mail System. It delivers to the **Destination**. It rarely checks if the **Sender** is telling the truth!

On-Path Attacks (The "Middle Man")

Concept: Passing Notes

Imagine Lucy passes a note to Linus in class.

- The Hacker Cat sits between them.
- He takes Lucy's note, reads it, **changes the words**, and passes it to Linus.
- Linus thinks the note came from Lucy.

This is an ****On-Path**** (formerly Man-in-the-Middle) attack.



Tools

Wireshark (to listen) and **Etercap** (to intercept).

ARP Poisoning: The Roll Call Lie

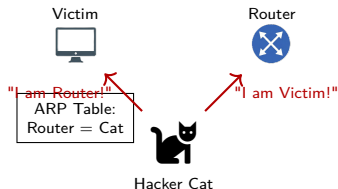
The Vulnerability

ARP is the protocol that finds MAC addresses ("Who has IP 10.1.1.1?").

- **Problem:** ARP has no ID check.
- **Analogy:** The teacher calls "Linus?", and the Bully yells "Here!" The teacher marks the Bully as Linus.

The Result

The Hacker Cat tells your computer "I am the Router." Your computer believes him and sends all your internet traffic to the Cat.

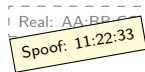


Tool Spotlight: MAC Spoofing

Changing Plates

Your MAC address is burned into the hardware, but software can override it.

- **Analogy:** Putting fake license plates on a getaway car.
- The camera (Switch/Router) sees the fake plate and lets the car through.



My Laptop



Looks like
a new PC!

Why do it?

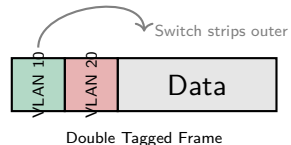
- 1 To bypass **MAC Filtering** (Access Control Lists).
- 2 To hide the device manufacturer (OUI) so hackers don't know what you are using.

VLAN Hopping (Double Tagging)

The Hidden Suitcase

How do you sneak a forbidden item past a checkpoint?

- **Double Tagging:** Put a packet inside a packet.
- **Outer Tag (VLAN 10):** "This is for the lobby." The first switch opens it.
- **Inner Tag (VLAN 20):** "This is for the Secure Vault." The switch sees this *after* opening the first one and sends it to the vault!



Case Study: The Coffee Shop Interception

Scenario: Free Wi-Fi Danger

Agent Snoopy is at the "KibbleKafe." He connects to the open Wi-Fi to check his bank.

The Hacker Cat is sitting in the corner. Suddenly, Snoopy's browser warns:

"Certificate Error: The identity of bank.com cannot be verified."

Snoopy realizes his traffic is being detoured through the Cat's laptop.

Detective Questions:

- 1 What Layer 2 protocol is the Hacker Cat abusing to redirect the traffic?
- 2 What is the Hacker Cat's position called (sitting between Snoopy and the Router)?
- 3 Why did Snoopy see a Certificate Error (HTTPS)?



Case Study Solution: The Coffee Shop Interception

Analysis

- 1 **Protocol: ARP.** The Cat used ARP Poisoning to tell Snoopy "I am the Router."
- 2 **Position: On-Path (Man-in-the-Middle).**
- 3 **Error:** The Cat tried to read encrypted (HTTPS) traffic. Since he doesn't have the Bank's private key, he used a fake certificate. Snoopy's browser detected the fake.

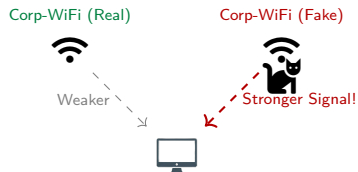


9.4 Rogue Devices and Evil Twins

Rogue Devices

Unauthorized hardware plugged into the network.

- Employee brings a home router to get better Wi-Fi.
- Hacker hides a "Raspberry Pi" behind a printer.



The Evil Twin (Wi-Fi)

The "Fake Starbucks" Attack.

- Attacker sets up a Wi-Fi spot with the **exact same name** (SSID) as the real one.
- Your phone sees two "KibbleCorp-WiFi" networks.
- It connects to the **strongest signal** (the attacker), giving them your data.

Rogue DHCP Servers

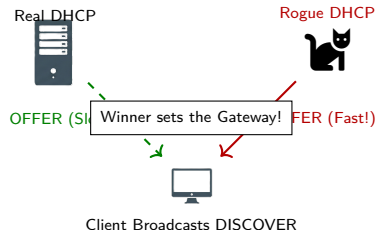
The Danger

If an attacker plugs in a router, it might start handing out IP addresses.

The "Race" Condition

Clients accept the **first** DHCP OFFER they receive.

- **Real DHCP:** "Here is IP 10.1.1.5, Gateway is Router."
- **Rogue DHCP:** "Here is IP 10.1.1.5, Gateway is **ME (The Cat)**."
- **Result:** All internet traffic detours through the attacker.



Defending with DHCP Snooping

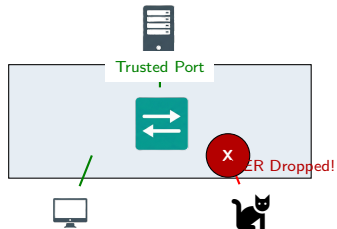
The Solution: DHCP Snooping

A security feature on switches that acts like a **Club Bouncer**.

Trust Boundaries

- **Trusted Ports:** Uplinks to real servers. *"VIPs allowed to speak."*
- **Untrusted Ports:** Regular user ports. *"You can listen, but you cannot hand out IPs."*

If an OFFER packet comes from an Untrusted port, the switch **drops it**.



DNS Attacks: Poisoning the Phonebook

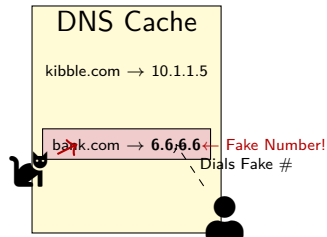
Concept: The Phonebook

DNS turns Names (google.com) into Numbers (8.8.8.8).

DNS Poisoning

"Changing the entry in the book."

- Attacker hacks the DNS Server.
- Changes "BankOfAmerica.com" to the Attacker's IP.
- You type the correct URL, but your browser goes to the **fake site**.



9.5 Social Engineering: Hacking the Human

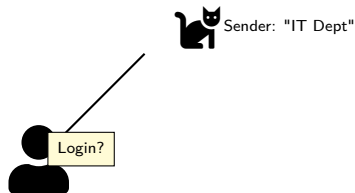
Phishing (The Bait)

Sending fraudulent messages to trick users.

- **Phishing:** Regular email scams ("You won a prize!").
- **Spear Phishing:** Targeted ("Hi Linus, this is your boss...").
- **Whaling:** Targeting the CEO/Big Fish.

Other Variants

- **Vishing:** Voice/Phone (Scam calls).
- **Smishing:** SMS/Text messages.



Example: Phishing Email Breakdown

A phishing message usually works because it looks urgent and slightly familiar. Teach students to slow down and inspect it.

Fake message

From: it-support@kibblecorp-security.com

Subject: Password Expiring Today

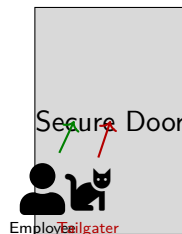
Click here to keep your account active: <http://kibblecorp-login.example/renew>

Red flags to spot

- Sender address is close, but not exact.
- The message creates urgency.
- The link uses plain HTTP and a suspicious domain.

Techniques

- **Tailgating:** Following an authorized person through a secure door ("Hold the door, please!").
- **Dumpster Diving:** Searching trash for sticky notes with passwords.
- **Shoulder Surfing:** Watching someone type their PIN or password.



Principles of Influence

Why does it work?

Attackers exploit human psychology ("Hacking the Human").

- **Authority:** "I am the CEO/Police, do this now!"
- **Intimidation:** "Do it or you're fired."
- **Consensus:** "Everyone else is doing it."
- **Scarcity:** "Only 2 iPhones left!"
- **Urgency:** "Account expires in 10 minutes!"
- **Trust:** "I know your friend Linus."

Stop. Look. Think.

Password Attacks

Brute Force

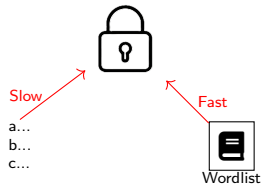
Trying **every possible combination** (aaaa, aaab...).

- *Analogy:* Trying every single key on a keychain until one fits.
- Guaranteed to work, but takes forever.

Dictionary Attack

Using a list of common words (dictionary).

- *Analogy:* Trying the most common house keys first.
- Fails if the password is complex ("Xy9#m!").



Tool Spotlight: John the Ripper & Salting

John the Ripper (JtR)

A tool used to test password strength by trying to crack them.

```
Loaded 3 hashes...
123456 (user1)
password (user2)
Session completed.
```

Defense: Salting

"Adding Spice to the Recipe."

- Before saving a password, we add random characters ("Salt").
- If Linus and Lucy both use "password123", their hashes will look completely different because they have different salts.
- Prevents pre-calculated attacks (Rainbow Tables).

Case Study: Intern Ike and the "CEO"

Scenario: 4:55 PM on a Friday

Intern Ike at KibbleCorp receives an email:

"From: ceo@kibble-corp.net (Notice the .net!)"

Subject: URGENT WIRE TRANSFER

I am stuck in a meeting. Wire \$5,000 to this vendor immediately. If this isn't done by 5:00 PM, it will be your fault. Do it now!"

Questions:

- 1 What specific type of social engineering is this?
- 2 Which "Principles of Influence" are being used?
- 3 What should Ike do?



Snoopy's Advice

- ❶ **Type: Whaling** (impersonating a big fish/exec) or Spear Phishing.
- ❷ **Principles:**
 - **Authority** ("I am the CEO").
 - **Urgency** ("By 5:00 PM").
 - **Intimidation** ("It will be your fault").
- ❸ **Action: Verify Out-of-Band.** Do not reply to the email. Call the CEO's assistant or walk to their office. Report to Snoopy immediately.

Recap: Offensive vs. Defensive

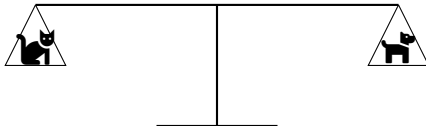
The Attacker (Red Team)

- **Spoofing:** Hiding identity (Masks).
- **Poisoning:** Redirecting traffic (Fake Signs).
- **Cracking:** Breaking passwords (Lockpicking).

The Defender (Blue Team)

- **Snooping:** Stopping rogue DHCP (Bouncers).
- **Honeypots:** Trapping intruders (Decoys).
- **Analysis:** Seeing the truth (Surveillance).

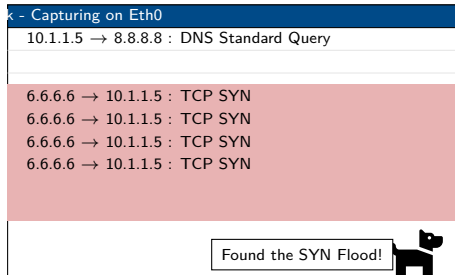
Security is a Balance



Analyzing the Attack with Wireshark

Red Flags

- **Flag 1:** Thousands of packets from one IP (DoS).
- **Flag 2:** Duplicate IP addresses (ARP Poisoning).
- **Flag 3:** Plaintext passwords (HTTP/Telnet).



Example: Incident Response Mini-Workflow

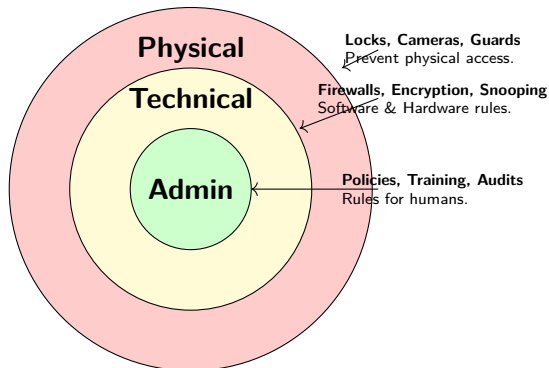
Security is not only about stopping attacks. It is also about what you do when something suspicious appears.

- 1 **Detect:** Notice the alert, log entry, or user report.
- 2 **Contain:** Disconnect the rogue device or isolate the host.
- 3 **Investigate:** Check logs, packet captures, and affected accounts.
- 4 **Recover:** Restore services from clean backups and re-enable access carefully.
- 5 **Improve:** Document the cause and update controls so it is less likely to happen again.

Simple classroom example

If a rogue DHCP server is discovered, the first move is containment: unplug the device or shut down the port, then verify clients received correct addresses.

Defense in Depth: Mitigation Strategies



No Single Solution

A firewall won't stop a user from holding the door open for a hacker. You need **all three layers** (Defense in Depth).

9.6 Additional Resources

Vulnerability Databases

- **CVE (MITRE)**: The dictionary of vulnerabilities.
- **NVD (NIST)**: Adds severity scores to CVEs.

Learning Tools

- **OWASP Top 10**: Web risks.
- **TryHackMe**: Legal practice labs.



Read the docs
before the
hackers do!

Knowledge is Power

Module 9.0 Summary I

Key Concepts:

- **CIA Triad:** **Confidentiality** (encryption, access control), **Integrity** (hashing, digital signatures), **Availability** (redundancy, DDoS mitigation).
- **AAA:** **Authentication** (who are you?), **Authorization** (what can you do?), **Accounting** (what did you do?). Use TACACS+ or RADIUS.
- **DoS/DDoS:** Denial of Service attacks flood targets with traffic. Mitigate with rate limiting, traffic filtering, and CDN protection.
- **Malware types:** Viruses (attach to files), Worms (self-replicate), Trojans (disguised), Ransomware (encrypt files for payment). Use antivirus, patching, and backups.
- **Social engineering:** Human-based attacks. **Phishing** (fake emails), **Vishing** (voice calls), **Smishing** (SMS), **Tailgating** (physical access). Defense: training and awareness.
- **Spoofing:** **MAC spoofing** (change NIC address), **IP spoofing** (forge source IP), **ARP poisoning** (man-in-the-middle), **DNS poisoning** (redirect to malicious sites).

Module 9.0 Summary II

- Rogue devices: **Rogue DHCP** (unauthorized IP assignment, VLAN hopping). **Rogue AP** (evil twin). Mitigate with DHCP snooping, 802.1X (port-based NAC), and wireless IDS.
- Password attacks: **Brute force** (all combinations), **Dictionary** (common words), **Rainbow tables** (precomputed hashes). Defense: complexity (8+ chars, mixed case, symbols), **salting**, **MFA**.
- **Defense in Depth**: Layered security. Physical security, network segmentation, firewalls, **IDS** (detect), **IPS** (prevent), ACLs, encryption (TLS, VPN), security training.
- **Wireshark**: Packet capture for forensics. Identify DoS floods, ARP poisoning (duplicate MACs), plaintext passwords (HTTP, Telnet). Filter by protocol, IP, port.

Security Operations:

- **Penetration testing:** Authorized simulated attacks (red team) to find vulnerabilities before attackers do.
- **Vulnerability scanning:** Automated tool (Nessus, OpenVAS) identifying known CVEs and misconfigurations.
- **Honeypots:** Decoy systems to lure attackers, gather intelligence, and distract from production assets.
- **SIEM:** Security Information and Event Management. Aggregate logs, correlate events, generate alerts. Examples: Splunk, QRadar.
- Security policies: Acceptable Use Policy (AUP), Incident Response Plan, Password Policy, Data Classification, Physical Access Controls.

Mission Accomplished!



Any questions?

Beginner Checklist: What to Turn On First

If you are securing a small lab or home office, start with controls that give the biggest benefit for the least complexity.

Highest Priority

- MFA for accounts that can reach sensitive systems.
- Automatic patching for endpoints and servers.
- Backups you have actually tested.

Next Layer

- Least privilege for users and admins.
- Logging and alerting on important systems.
- DHCP snooping, port security, and basic ACLs on the network edge.